Yumeng Cai

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EDUCATION

Georgia Institute of Technology (*Master's in Robotics)* Atlanta, GA, USA
08/2025-present

South China University of Technology (SCUT) Guangzhou, China
Bachelor of Engineering in Robotics Engineering GPA: 3.78/4.0 09/2021-06/2025

- Second Class Scholarship for Excellence (Zhuoyue) and Outstanding Student Union Leader for 2022-2023
- Third Class Scholarship for Excellence (Huameng), Merit Student, and Outstanding Student Leader for 2021-2022
- Third Class Scholarship for Excellence (Zhuoyue), Merit Student, and Outstanding Student Leader for 2023-2024

PATENTS & PAPERS

[Accepted Invention Patent] A Gesture Recognition Smart Bracelet System based on Surface Electromyography Signal (second patentee) 04/2023

[Utility Model Patent] A Base Station Energy Storage Box (second patentee) 03/2023

[Accepted Paper] *Y. Wang*†[1], ***Yumeng Cai***†[1], *M. Pang, Y. Xu, G. Kassab, C. Zhou, G. Chase, X. Chen, H. Luo and Y. Chen*[1*], **Fluid-Structure Interaction of Left Ventricle with Impaired Ventricular Contractility**, Acta Mechanica Sinica (AMS) 08/2025

[Accepted Paper] *B. Xie, Y. Wang,* ***Yumeng Cai****, Zi. Ouyang, C. Zhu, H. Sun, and Y. Chen*[1*], **Computational Fluid Dynamics Analysis of Tandem Floating Offshore Wind Turbines under Coupled Pitch-Surge Motion: A Comparative Study of the National Renewable Energy Laboratory 5 Megawatt and the International Energy Agency 22 Megawatt Turbines**, Physics of Fluids 03/2025

[Accepted Paper] *Y. Wang*†[1], ***Yumeng Cai***†[1], *B. Xie, C. Zhu, H. Luo, Y. Li, and Y. Chen*[1*], **A Computational Study of the Hydrodynamics of Quadrupedal Paddling**, Biomimetics 06/2024

[Accepted Paper] *L. Gan, Y. Xu,* ***Yumeng Cai****, Y. Zhou,* **A High Sensitivity Flexible Stretchable Fiber Optic Sensor based on the Waveguide Loss Principle**, IEEE Transactions on Instrumentation & Measurement 05/2024

[Accepted Paper] *J. Liang, G. Lou, F. Zhou,* ***Yumeng Cai,*** *C. Wang, Y. Zhou*[1*], **Residual Reinforcement Learning Based on Inverse Kinematic Modeling for Soft Robotic Arm Control**, ASME Journal of Dynamic Systems, Measurement, and Control 10/2024

[Accepted Paper] *H. Huang, S. Zhang, Y.Chen, S. Liu, Z. Chen,* ***Yumeng Cai****, H. Li, L. Xie Member. IEEE,* **Portable sEMG-Controlled Hand Electrical Stimulation System**, IEEE Sensors Journal 08/2024

SELECTED RESEARCH EXPERIENCE

Image-Guided Closed-Loop Robotic Guidewire Navigation System 09/2025-02/2026

- *Graduate Thesis Project; Advisor: Professor Jaydev P. Desai*

- Design and develop a two-way bending guidewire with control platform
- Develop a C-arm–based guidewire perception pipeline to robustly track the guidewire
- Design and implement real-time feedback control to autonomously steer the guidewire to targets

Soft Wearable Assistive Robots Exosuit 07/2024-09/2024

- *University of Notre Dame - Summer Research Program Project (iSURE);*
- *Advisor: Professor Margaret Coad*

- Researched soft and bioinspired designs, models, and control methods for exosuits and prosthetic limbs
- Took charge of improving the entire system, liaising with the health center, and collecting data to further make the wearable device more comfortable for users

A Six-Degree-of-Freedom Soft Continuum Robot Based on the FBG Principle for Shape Reconstruction in Minimally Invasive Surgery 01/2024-06/2025

- *Undergraduate Thesis Project*

- Designed innovative miniature continuum robots by integrating cable-driven and pneumatic actuation mechanisms

- Determined the current shape of the actuator by integrating data from FBG sensors with a model
- Achieved independent 6 degrees of freedom; the actuator's diameter is 10mm

Jellyfish Amphibious Robot based on Steady State Response 06/2023-09/2024

- Simulated soft actuators of various sizes and fabrication processes, and designed jellyfish-like structure
- Conducted mechanical performance and integrated testing of the soft actuator
- Precisely controlled all modules, including circuits and sensors through a microcontroller; established the system control logic and improved the execution accuracy
- Engaged in data collection, analysis, and drafting of the thesis; expected to be completed by the beginning of 2025

A Computational Study of the Hydrodynamics of Quadrupedal Paddling– Physics of Fluids 09/2023-07/2024

- *Research Member in a Team of 3 Students*
- Calculated the forces, power consumption, and impact on flow fields for both single-leg and quadruped configurations across various gait patterns, including diagonal and lateral sequence gaits, along with the initial swing angle, swing range, and power phase of the legs in each gait
- Processed large volumes of calculated data with Ubuntu Subsystem using a four-legged robot dog model
- Made realistic renderings of fluids and solids with Tecplot

A High Sensitivity Flexible Stretchable Fiber Optic Sensor based on the Waveguide Loss Principle - IEEE Transactions on Instrumentation & Measurement 01/2023-06/2024

- *Research Member in a Team of 3 Students*
- Fabricated a flexible, stretchable fiber optic sensor optimized for continuous health monitoring
- Conducted a series of performance experiments to characterize the sensor's ability to measure large strains, low detection limit, and high repeatability
- Made a Human-Machine Interaction (HMI) system to translate movements into signals for interaction

Gesture Recognition Smart Bracelet with Low Delay and Low Power Consumption based on Surface Electromyography Signal 07/2022-05/2023

- Developed solutions for various real-world problems, including: (1) detecting the weak sEMG signal on the wrist and overcoming the noise interference; (2) offline computing on the cloud end; (3) the decreased accuracy of gesture recognition due to changes in wearing style and individual differences
- Employed array-based sEMG bracelets, adaptive recognition technology, edge computing, etc.; utilized the fine-tuning model to enhance the bracelet's robustness
- Established a deeper understanding of designing PCB layout; operated and deployed cloud training models, and constructed a CNN artificial intelligence network

COMPETITIONS

Third Prize Winner, Basic Theory and Key Technology Seminar and Soft Robot Innovation Design Competition, The 9th Soft Robotics Conference 11/2024

Third Prize Winner, 2023 National College Students Electronic Design Competition 08/2023

Third Prize Winner**,** The 22nd Robocon National College Student Robot Competition 07/2023

Person in charge and Holder of the Winning Prize**,** The 9th China International College Students "Internet+" Innovation and Entrepreneurship Competition 06/2023

Person in Charge and Special Prize Winner**,** 17th Guangdong Xinwanda "Challenge Cup" College Students Extracurricular Academic Works Competition 05/2023

Third Prize Winner**,** College Students Asia and Pacific Mathematical Contest in Modeling 01/2023

SKILLS & OTHERS

Language	English (Proficient; TOEFL:100, Speaking: 23; Writing: 28), Chinese (Native)
Technical Skills	• Programming Languages: C, C++, Python, R • Tools: MATLAB, Solidworks, Pycharm, Altium Designer, Visual Studio, Arduino IDE, CubeMX, Ubuntu Subsystem
Interests	Draughts, Tai Chi (Chinese Kungfu), Soccer, Dundubhi (Chinese red drum), Calligraphy
Additional Experience	"Machine Learning Winter Program" organized by Girton College, Cambridge. Supervised by Prof. Pietro Li*o*, got 69% grade on the UK scale